

## Synthesis Gas:

Raw materials for 1st generation Petrochemicals.  
→ Synthesis gas ( $\text{CO} + \text{H}_2$ ) obtained from steam reforming of hydrocarbon.

→ Olefins from pyrolysis of hydrocarbon

→ Aromatic from catalytic reforming of gasoline.

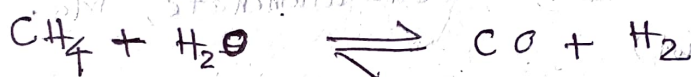
### 1. Synthesis Gas ( $\text{CO} + \text{H}_2$ )

Partial oxidation of  $\text{CH}_4$

Ratio  
 $\text{H}_2 : (\text{CO} + \text{CO}_2)$

Aim: to get the suitable ratio

Steam reforming reaction

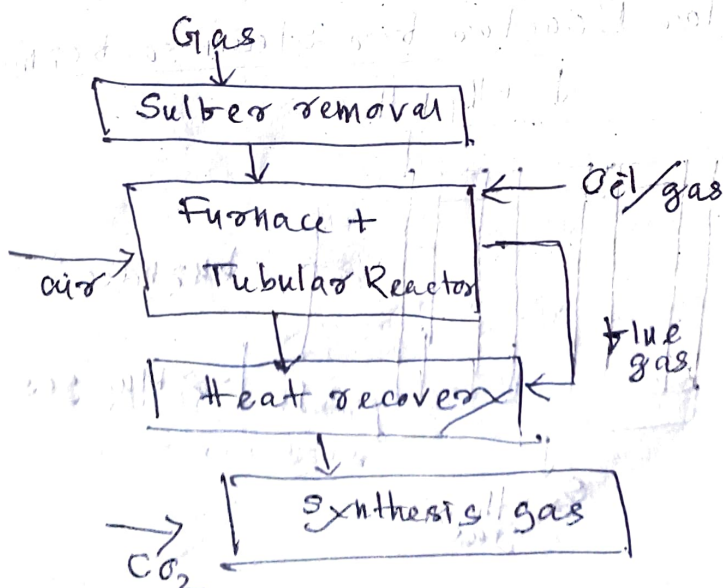


$$\Delta H = +49 \text{ kcal/mol}$$



water-gas shift reaction  
/ shift reaction conversion.

Flow diagram:



# Composition of Synthesis gas from methane steam reforming

## Components:

CO

Vol%

CO<sub>2</sub>

8.1

H<sub>2</sub>

75.7

N<sub>2</sub>

0.2

CH<sub>4</sub>

0.5

H<sub>2</sub>S

0

## Catalyst:

Ni/Al<sub>2</sub>O<sub>3</sub>

Ca-aluminate, Mg-aluminate

(6-20% wt)

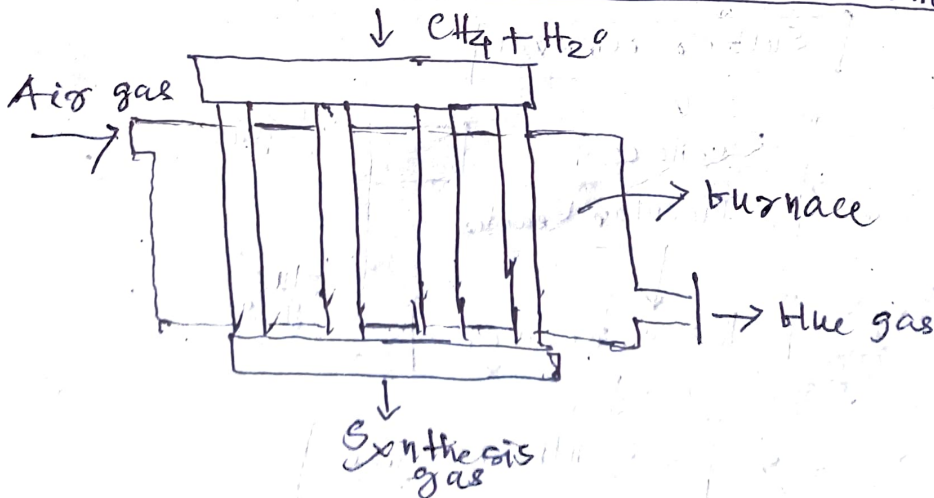
H<sub>2</sub>: CO<sub>2</sub> = 3:1

## Shift Conversion



(Iron-chrome catalyst)  
300°C - 500°C

## Tubular Reactor for steam reforming:

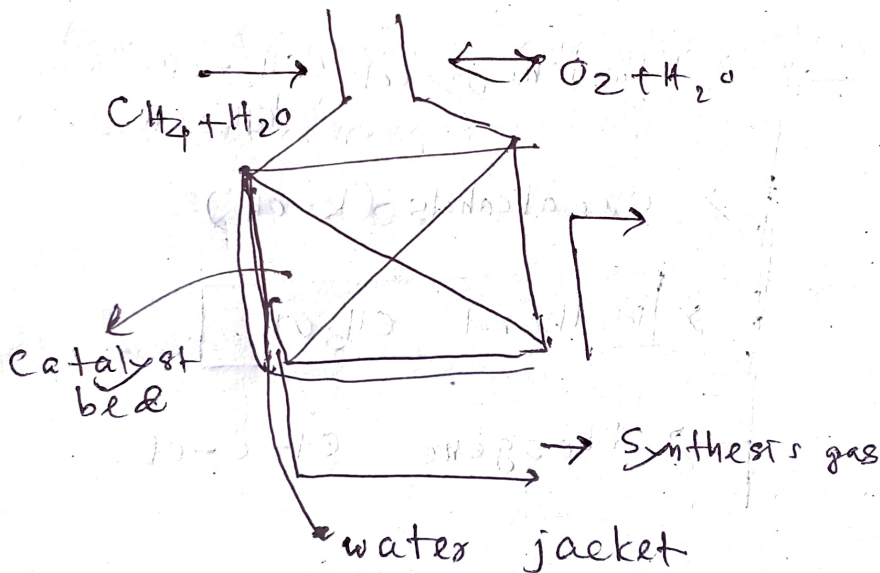


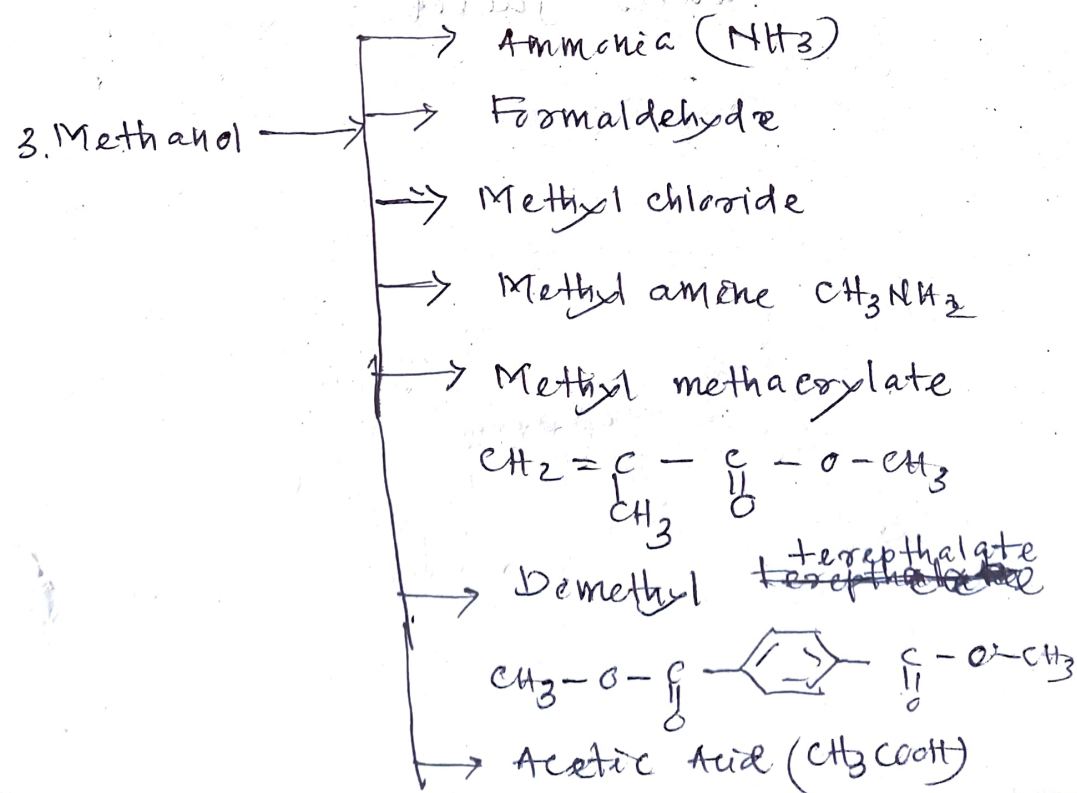
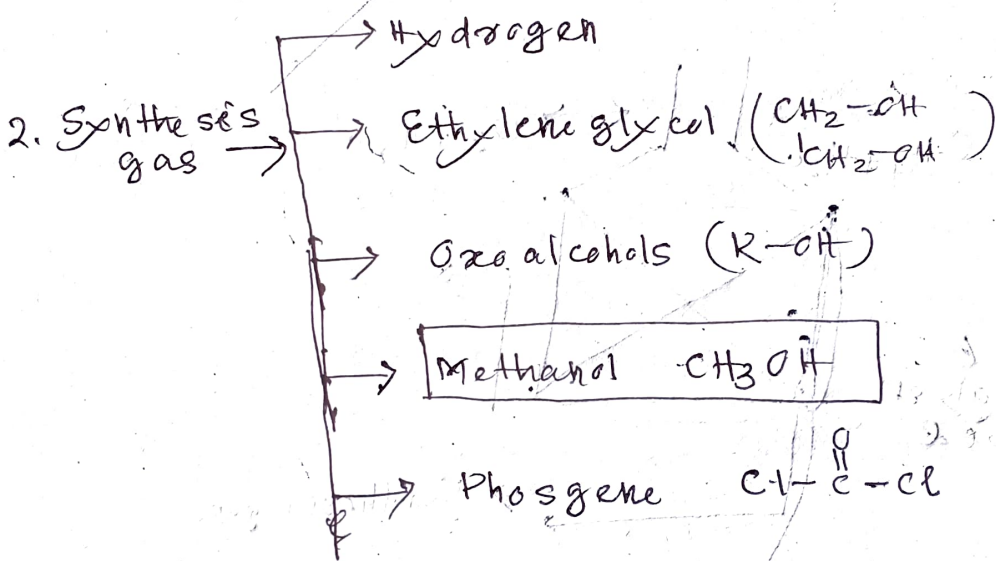
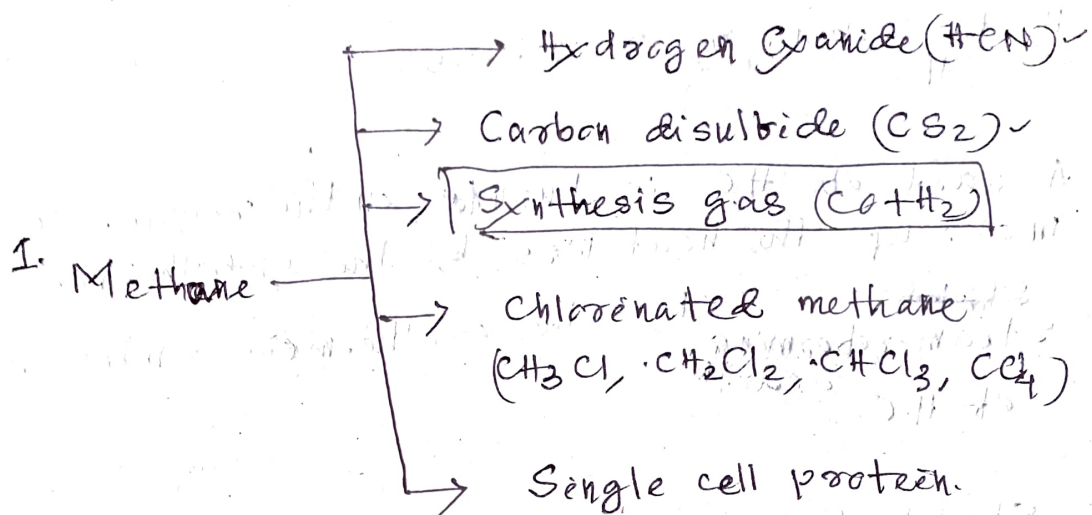
# Autothermal Reaction / Oxidative process

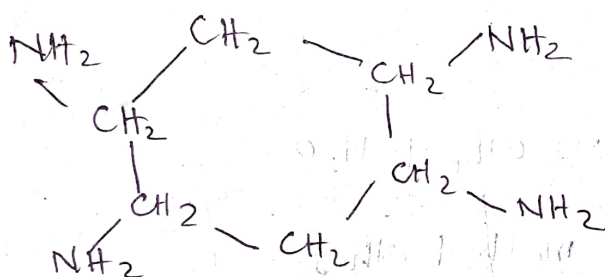
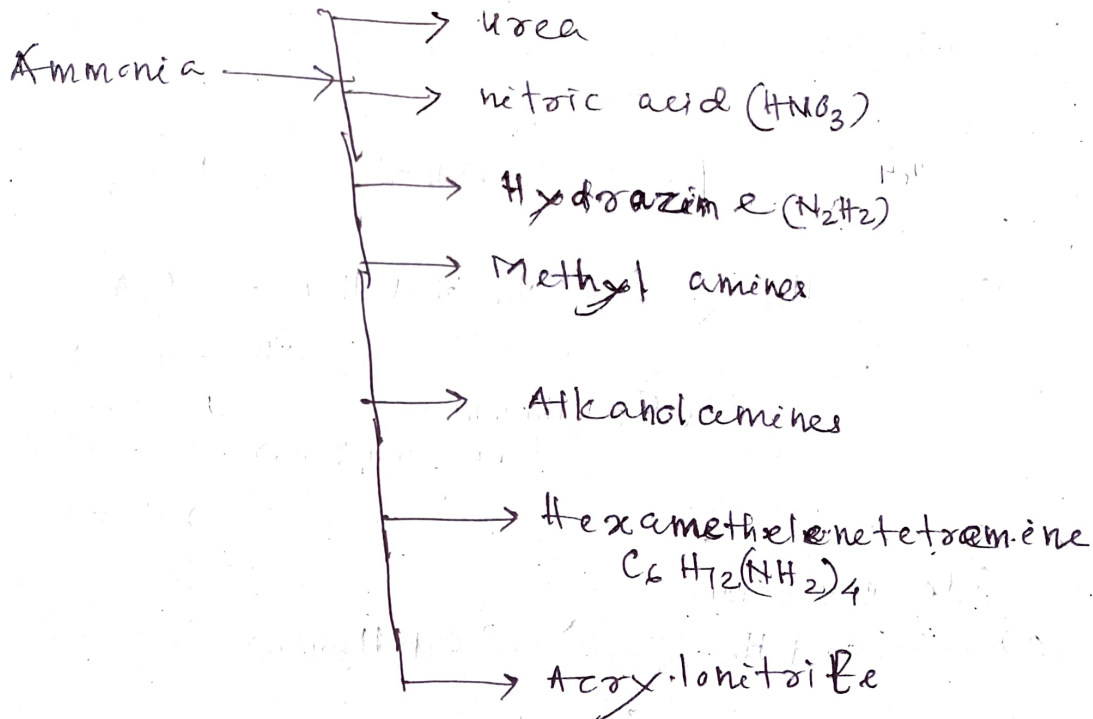
Hydrocarbon:  $O_2 \approx 1:0.55$   
(H:C)

A part of H.C is burnt in the reactor to make-up the heat loss by the endothermic ~~Shift~~ Steam-reforming reaction by the exothermic combustion of H.C.

## Shift Continuous reactor

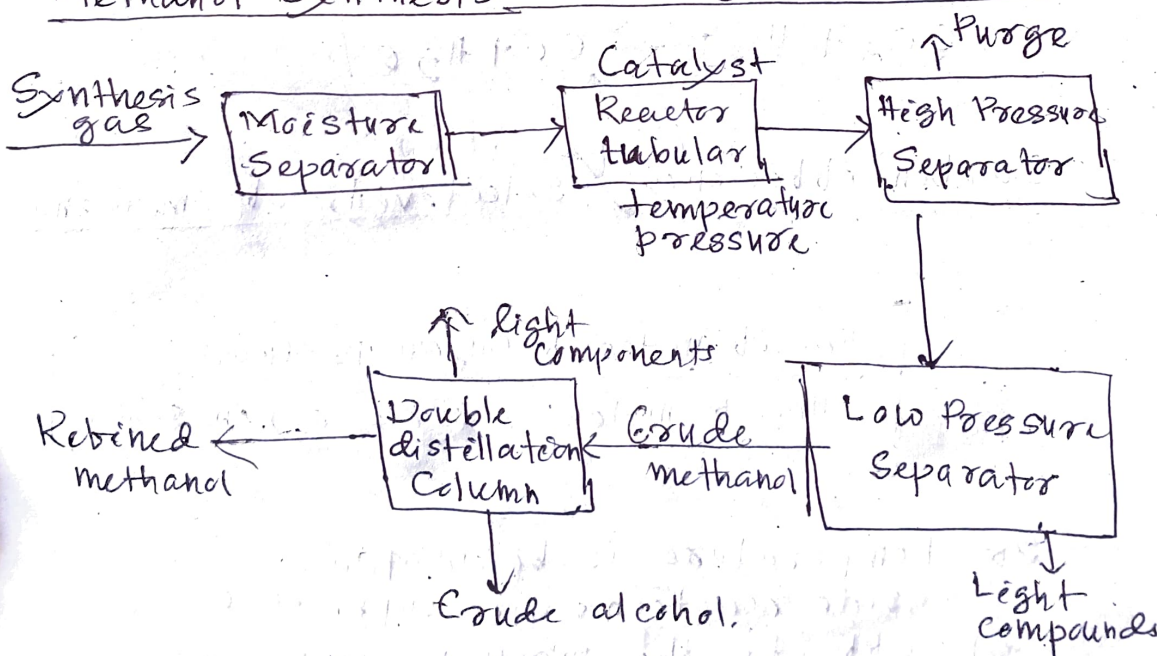






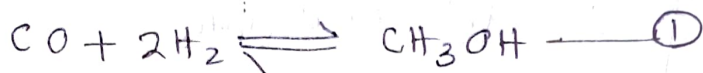
Hexamethylenetetramine.

### Methanol Synthesis: from Synthesis Gas:

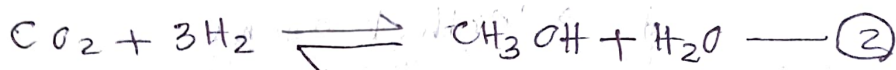




## Reactions:



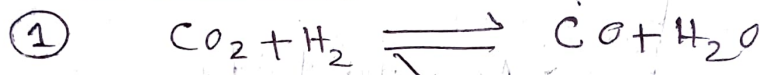
$$\Delta H = -90.8 \text{ kJ/mol}$$



$$\Delta H = -49.5 \text{ kJ/mol}$$

(Exothermic Reaction)

### Side reaction

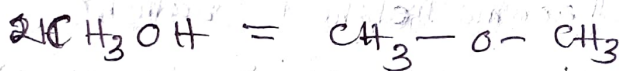


Endothermic reaction

### (2) Methanation:



### (3) Formation of methyl ether



Carbon efficiency / selectivity of methanol:

$$= \left[ \frac{\text{no. of moles of CH}_3\text{OH produced}}{\text{no. of moles of (C} + \text{CO}_2 \text{) in gas}} \right] \times 100$$

→ If temperature is below  $400^\circ\text{C}$

side reactions are suppressed but it affects the main process, like reaction becomes very slow, and cost increases.

Production of methanol is favoured by

1. Elevation of Pressure
2. Reduction of temperature
3. Increase of  $\text{CO}/\text{CO}_2$  ratio.
4. Increase of  $\text{H}_2$  content in syn gas.

→ Copper oxide insertion in the catalyst can give the optimum condition.

But it can be abetted by the impurities and degraded with time.

→ Thorium and Copper ( $\text{Th Cu}_6$ ) are melted together to give  $(\text{Th Cu}_6)$  which is 10 times more active than the copper catalyst.

- It decreases the side reactions

- It is very stable and robust catalyst

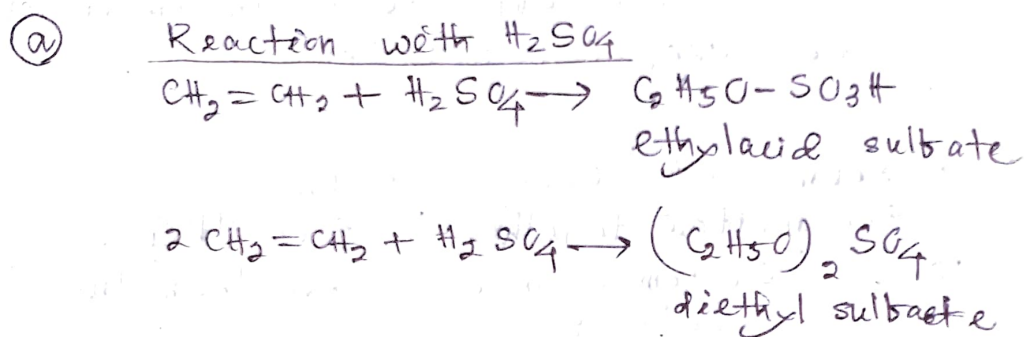
→ Methyl methacrylate

Anti-knock agent for gasoline

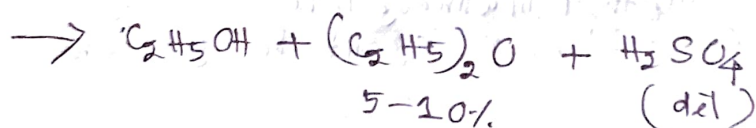
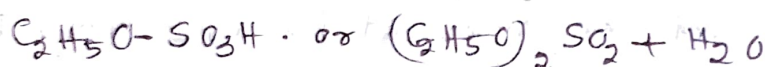
# Ethanol:

## Preparation Method

1. Fermentation process for agricultural wastes
2. Synthesis from ethylene.
  - (a) Indirect hydration of ethylene
  - (b) Direct hydration of ethylene.



### Hydrolysis:



Reconcentration of dil.  $H_2SO_4$  by  
evaporation under vacuum.

Condition of reaction in absorber:

Feed  $\rightarrow$  min 35 vol% ethylene +  
(methane + ethane)

Counter current operation in  
98%  $H_2SO_4$  in absorber

Temp:  $80^\circ C$

Pressure:  $1.3 \times 10^6 \text{ Pa} - 1.5 \times 10^6 \text{ Pa}$

$\rightarrow$  Increasing temperature must be avoided  
to avoid the resin formation.



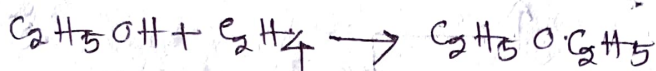
### Limitations

- Corrosiveness of concentrated  $H_2SO_4$
- Process is costly.
- High production of diethyl ether.

### (b) Direct hydration of ethylene:



→ side reactions  $\Delta H = -43.4 \text{ kJ}$



- Pressure should be maintained in such a way that product is more and polymerization of ethylene is avoided.

ethylene : deionized water

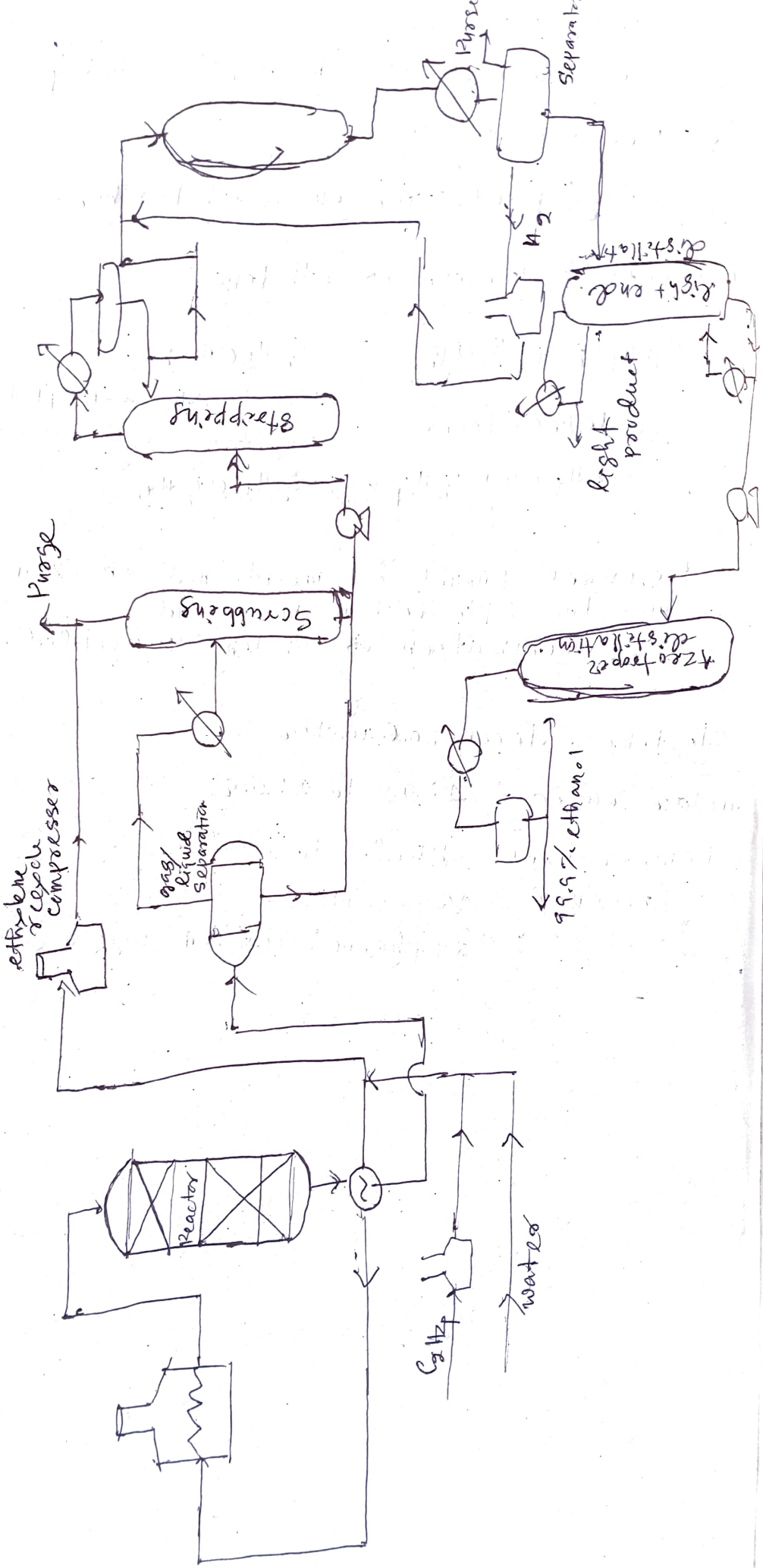
molar ratio = 1:0.3 to 1:0.8

temperature =  $250^\circ\text{C}$  to  $300^\circ\text{C}$

Pressure = 6 - 8 MPa

Catalyst = Phosphoric acid on silica ( $SiO_2$ )

Process Flow Diagram:



## Ethanol:

55-65 trays are required

Benzene is used as the solvent  
for 95% ethanol-water mixture  
Azeotropic distillation.

From the bottom  $\rightarrow$  99.9% ethanol

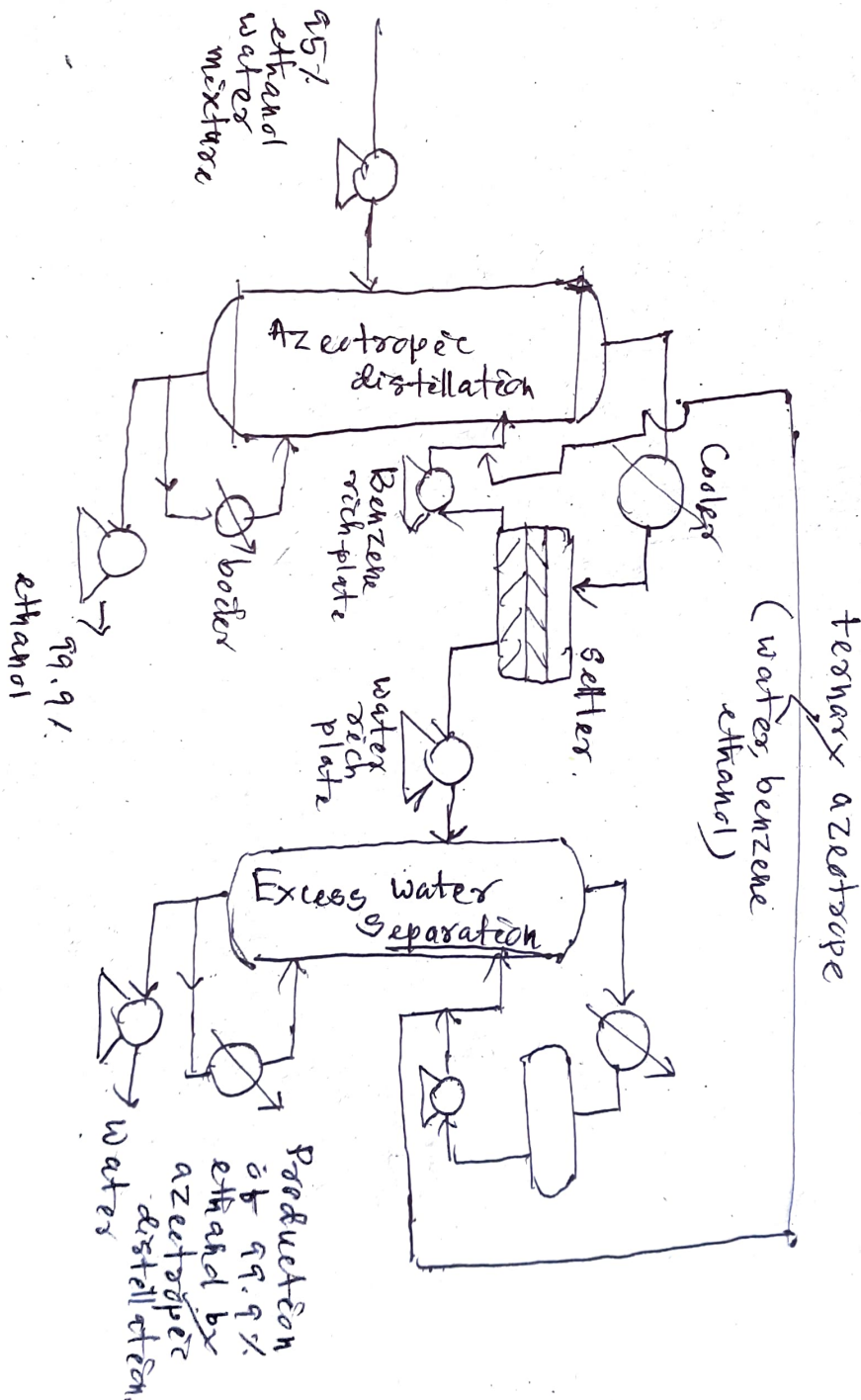
top - ternary mixture

(benzene, water, ethanol)

Usage:  $\rightarrow$  As a fuel

$\rightarrow$  to increase the octane number

$\rightarrow$  As an additive for fuel.



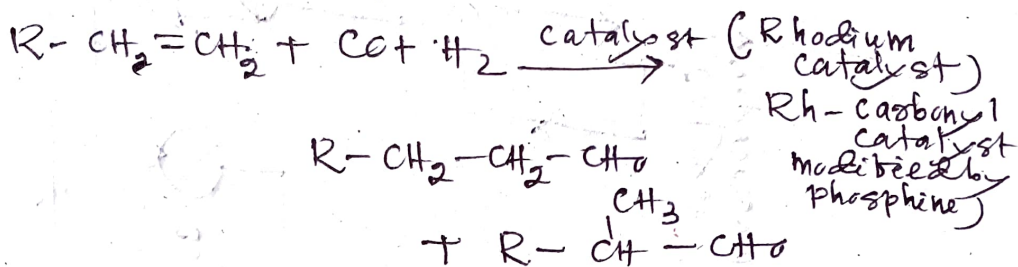
## Higher Alcohols:

C<sub>3</sub> - C<sub>20</sub>

Olefin + Synthesis gas  $\rightarrow$  Aldehyde  
(having more than one carbon atom than the olefin)

## Oxo Process:

### Hydrobormylation Reaction:



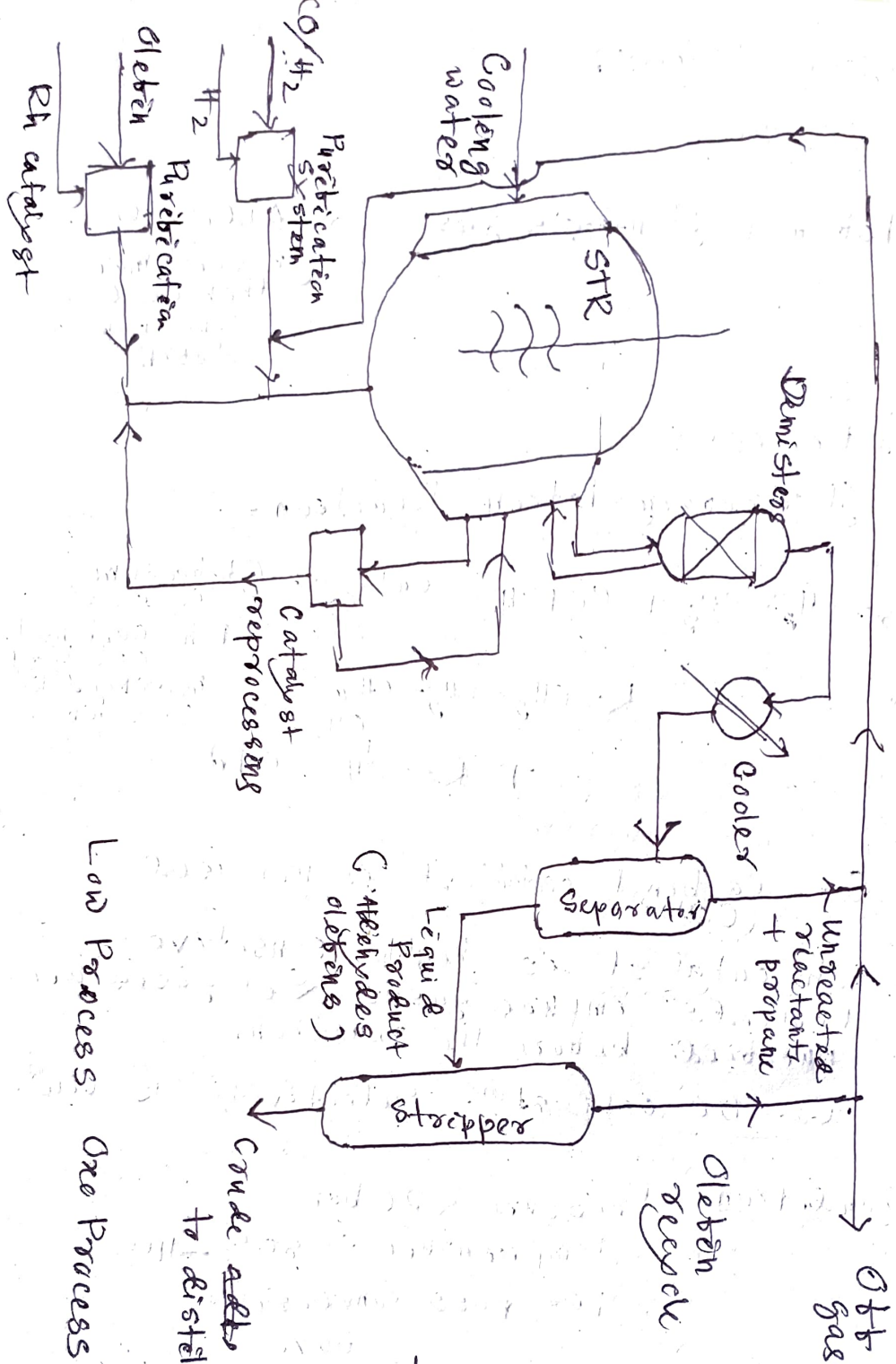
- \* Co-Carbonyl catalyst is not used
- \* Rh-catalyst is highly sensitive towards impure gases. So, gases are purified before the reaction.
- \* Co-CO catalyst: Selectivity is low.

### Condition:

Pressure < 20 bar  
Temperature  $\pm$  85°C - 115°C  
per pass conversion  
30%.



# Flow Diagram;



Low Process One Process (LP)

STR  
→ stirred Tank  
Reactor

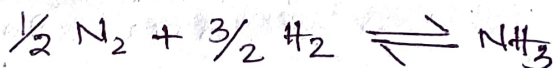
Can be ~~also~~ alcohol  
to distillation

## Oxidation of paraffins to higher alcohols:

- produces not only alcohols but also ketones and aldehydes.
- It is non-selective
- Catalyst: ~~Perm~~ Boric Acid
- Paraffin + air  $\xrightarrow[0.1\% \text{ KMnO}_4 + 5 \text{ wt\% boric acid}]{150-200^\circ\text{C}}$  70 mole% alcohol + 20% ketones + 10% organic acids
- Reaction is un-selective.
- Catalyst is recycled back to the blow.

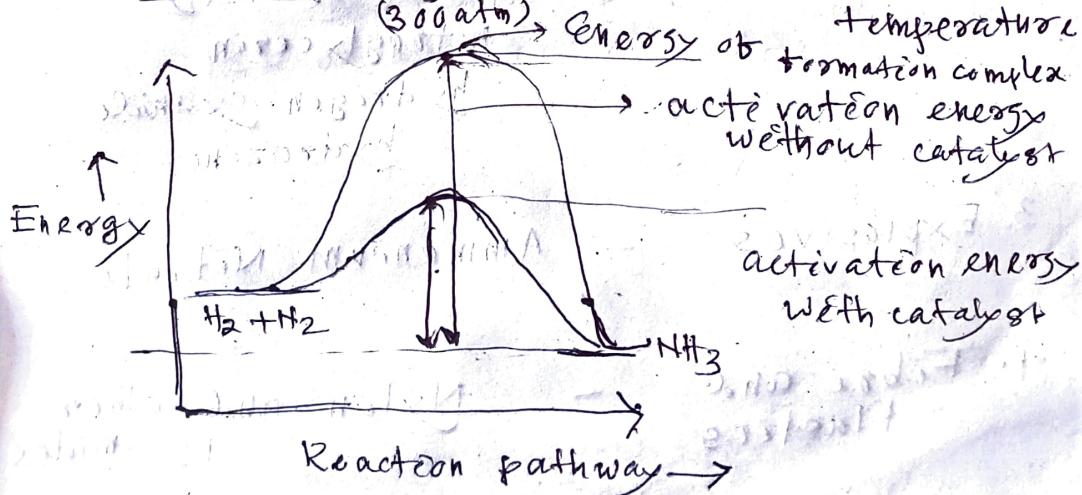
## Preparation of Ammonia (NH<sub>3</sub>):

Haber process:



exothermic reaction.

Condition: High pressure and low temperature (300 atm)



$$K_{eq} = \frac{P_{NH_3}}{P_{N_2}^{1/2} P_{H_2}^{3/2}}$$

### Catalyst:

Iron oxide with oxides of aluminium, (Al<sub>2</sub>O<sub>3</sub>) zirconium or silicon at about 3% (ZrO<sub>2</sub>) (SiO<sub>2</sub>)

+ Potassium oxide at about 1%

### Uses of Ammonia:

#### Industry

##### 1. Fertilizer

For production of ammonium sulphate (NH<sub>4</sub>)<sub>2</sub>SO<sub>4</sub>

Ammonium phosphate

Ammonium nitrate  
urea

##### 2. Chemicals

Synthesis of nitric acid  
(for preparation of TNT)

nitroglycerin

hydrogen cyanide

hydrazine

##### 3. Explosives

Ammonium Nitrate

##### 4. Fibre and Plastics

Nylon and other polyamides

##### 5. Refrigeration

for making ice,  
large scale refrigeration plant,  
air conditioning units

6. Pharmaceuticals - Manufacture of drugs,  
Sulphonamide,  
anti-malaria drugs  
Vitamin B: nicotinamide,  
thiamine

7. Pulp and paper - Ammonium bisulfate

8. Meneng and metallurgy - Nitriding  
(bright annealing)  
of steel,  
used in zinc  
and nickel extraction

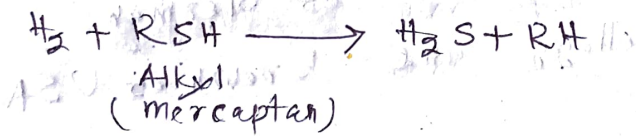
new catalyst instead of iron oxide  
Ru/graphite.

→ The pressure can be reduced to  
minimize the cost.

Ammonia Production:

Natural gas ( $\text{CH}_4$ ) feedstock: -  $P = 25-35 \text{ bar}$

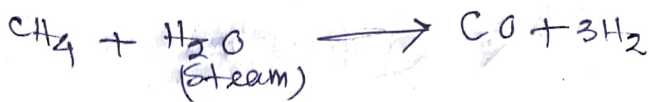
1. Sulfur removal from feedstock: -



2.  $\text{H}_2\text{S(g)}$  removal:

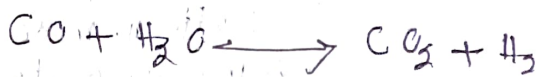


3. Steam reforming (catalytic)



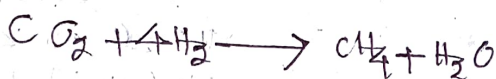


#### 4. Shift Conversion:

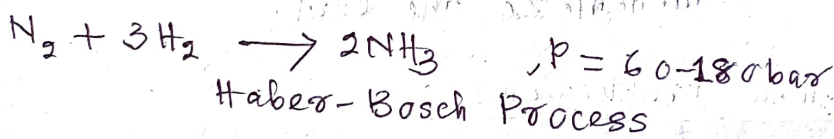


→  $\text{CO}_2$  is removed either by absorption in ethanal-amine solution or adsorption by pressure-swing adsorption.

#### 5. To remove CO and $\text{CO}_2$



↓  
Methanation reaction  
to remove any  
residual CO or  $\text{CO}_2$ .

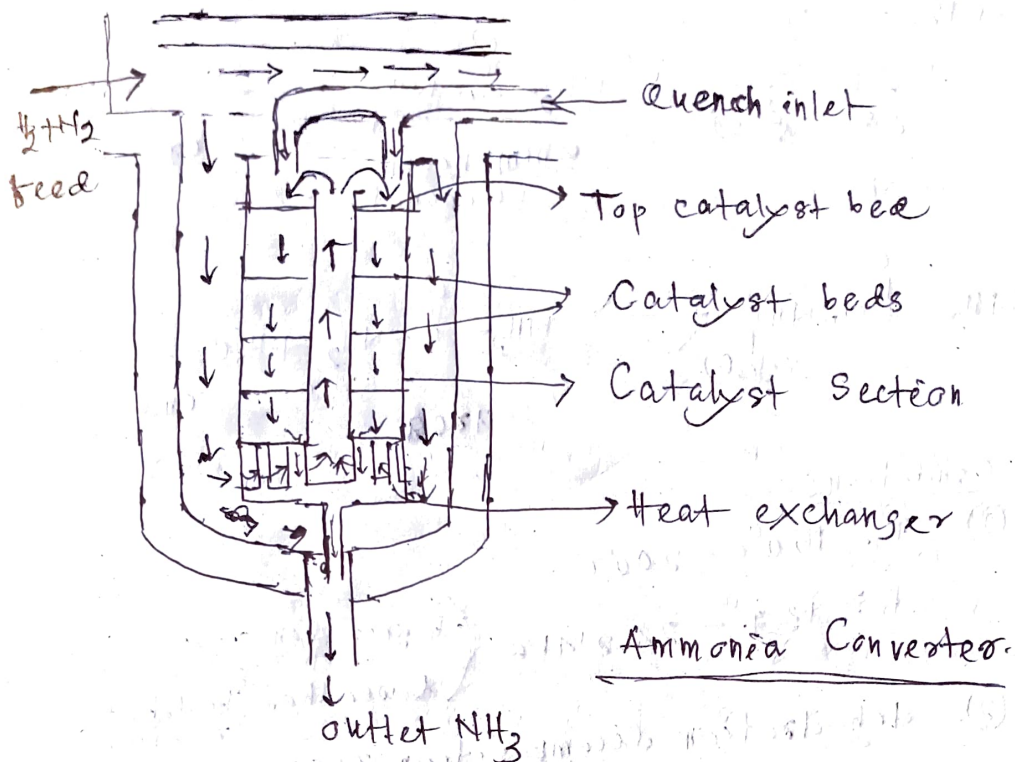


→ Halowax - Topsoe of Denmark

→ Uhde - GmbH of Germany

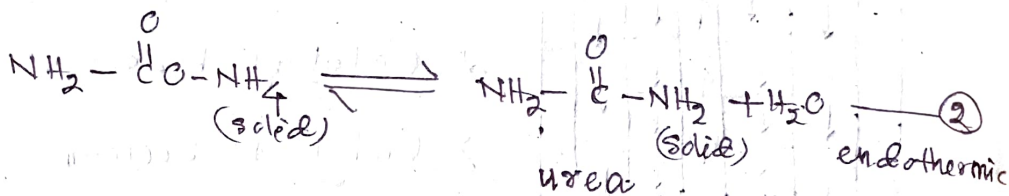
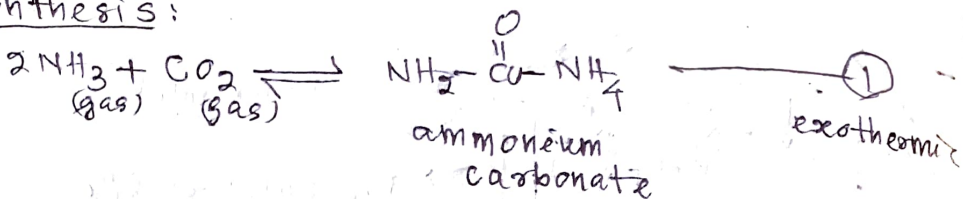
→ Kellogg and Brown of USA

# Diagram:



# Urea

## Synthesis:



## Conditions:

①  $T: 150^\circ\text{C} - 200^\circ\text{C}$

$P: 13.7 - 23.2 \text{ MPa}$  (higher than vapor pressure of  $\text{NH}_2\text{C}(=\text{O})\text{NH}_2$   
lower than the dehydration reaction)

② dehydration decomposition reaction.

Preparation of solid urea  $\rightarrow$  from molten urea

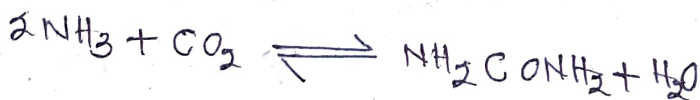
1. Prilling  $\rightarrow$  (a) fluidized  
2. Granulating (better)  $\rightarrow$  (b) non-fluidized

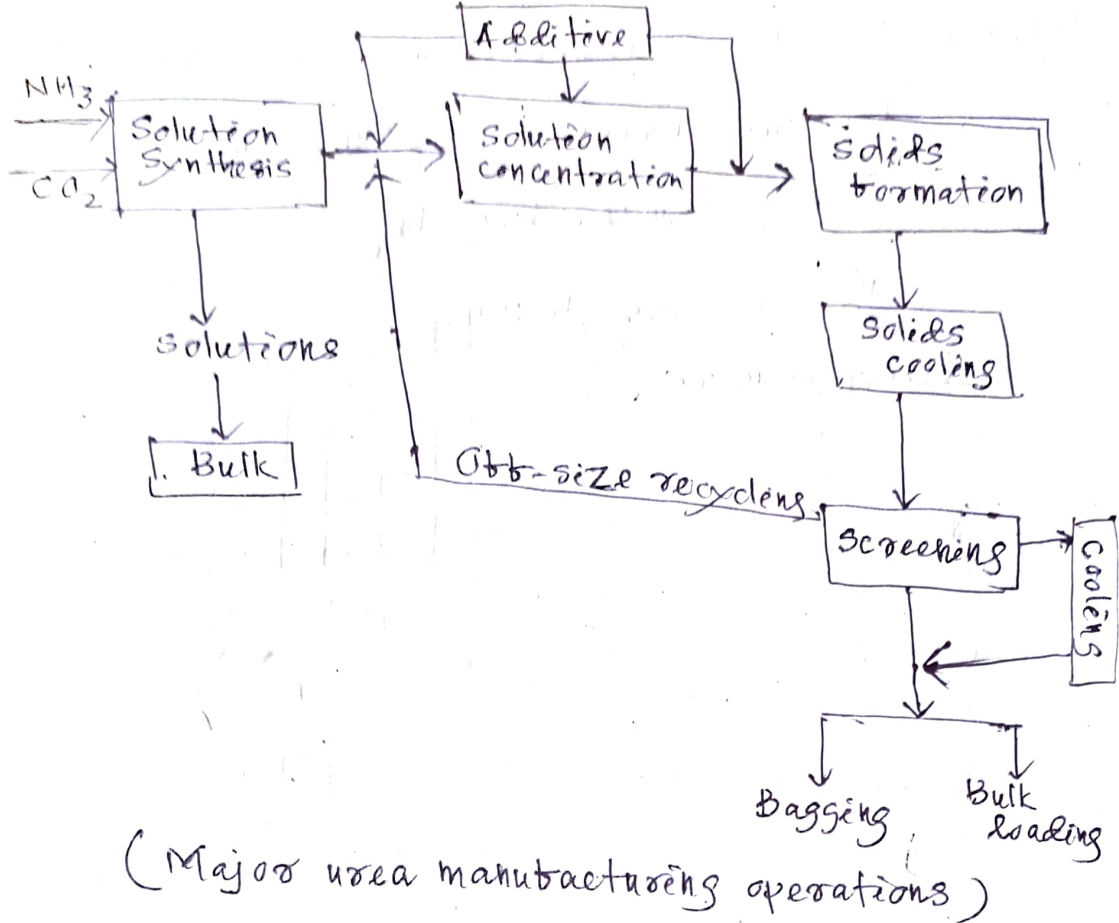
$\rightarrow$  Drum granulation (better)  
 $\rightarrow$  Pan granulation

$\rightarrow$  maximum conversion:  $T = 180^\circ\text{C}$   
Using more  $\text{NH}_3$

$\rightarrow$   $\text{NH}_3$  and  $\text{CO}_2$  come from the same plant.

## Overall reaction:





### Once-through Urea Process:

- No recycling
  - Output of the process is absorbed in dilute acids.   
 (unreacted  $\text{NH}_3$ , unconverted  $\text{NH}_2\text{-COONH}_4$ )
- After the urea is separated.

### Solution-recycle Process:



# UREA-(DSM) - Stamicarbon BV Process

Reactor pressure: 130-150 atm

Temperature: 170-190°C

$\text{NH}_3 : \text{CO} = 2.8 \text{ to } 2.9$

Stripping unit:

# Water conditioning / water treatment:

1. Surface water  $\rightarrow T$
2. Ground water

## 1. Surface water

- (i) Temperature changes
- (ii) Contains less minerals
- (iii) Mostly used in Industries
- (iv) Water on the surface of earth  
(Sea, ponds, lakes...)

## 2. Ground water

- (i) Remains constant
- (ii) Contains heavy amount of minerals.
- (iii) not preferable to use in industries
- (iv) Water stored inside earth.
- (v) It is softer than ground water (hard water)  
hard water containing dissolved salts of Ca and Mg.  
(bicarbonate, chloride, sulphates, nitrates etc. with low thermal conductivity)  
expensive to use  
(causes scale)
- (vi) less costly to use

Hardness: It is usually expressed in term of the Dissolved calcium and magnesium salts as calcium carbonate equivalent.

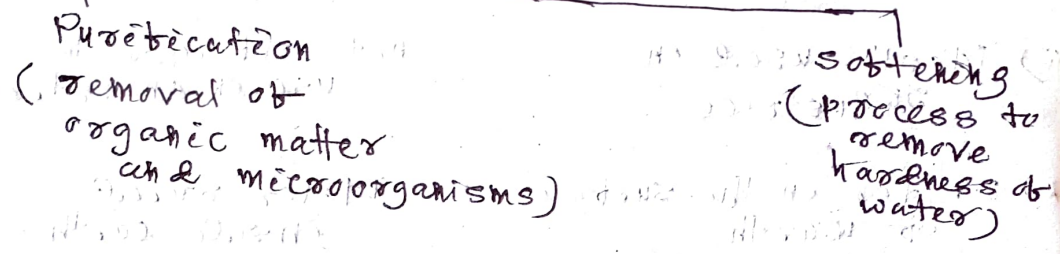
- (i) Carbonate or temporary hardness  
(due to bicarbonate salts of Ca/Mg)
- (ii) non-carbonate or permanent hardness.  
(due to sulphates, chloride of Ca and Mg)

TDS - total dissolved solids

Other impurities presents are suspended insoluble matter, organic matter, color, and dissolved gases. (true  $\text{CO}_2$ ,  $\text{O}_2$ ,  $\text{N}_2$ ,  $\text{Si}$  in water  $\text{H}_2\text{S}$ )

units of expressing water analysis  
 parts per million (ppm)  
 → In 1 litre how many  $\text{mg}$  are dissolved  
 grains per gallon (gr/gallon),  $\text{mg/l}$

## Water Conditioning



### Water Softening:

#### (1) Ion exchange:

It is a chemical reaction where the mobile ions of the ion exchangers are exchanged with the hard ions of water equivalent to equivalent likes to likes.

#### → Cation exchange:

• Na-styrene divinyl benzene sulfonate (Na-resin)

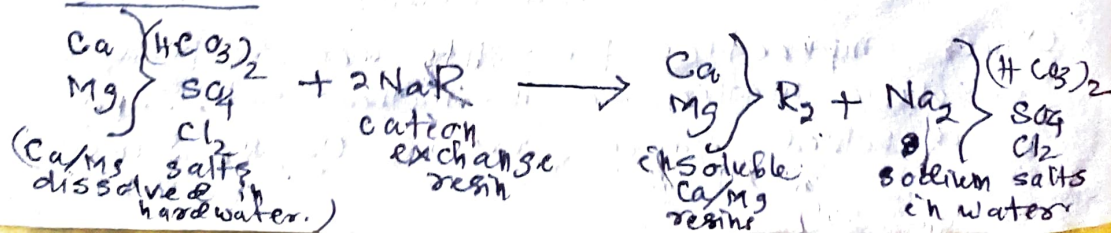
• zeolite (sodium aluminosilicate,  $\text{NaAlSi}_3\text{O}_8$ )

#### Regeneration of resins

treated with brine solution (NaCl) to remove the Ca and Mg ions from the saturated resin.

Reaction condition:  $T \sim 150^\circ\text{C}$   
 $\text{pH} \sim 9-14$

#### Reaction



→ Anion exchange:

It is the demineralization of negative ions like  $\text{Cl}^-$ ,  $\text{SO}_4^{2-}$  etc. from water.